Physics 121 Equations and Constants Sheet

Constants & Units

C = Coulomb = Charge = q
e = electron charge = -1.602×10^{-19} C
$\epsilon_0 = 8.85\times10^{-12}\ \frac{C^2}{Nm^2}$
$K_e = \frac{1}{4\pi\epsilon_0} = 9\times10^9\ \frac{Nm^2}{C^2}$
J = joule = energy
W = work = J
eV = electron Volt = 1.602×10^{-19} J
$m_e = 9.109\times10^{-31}$ kg = mass of electron

F = Force = Newtons = N
t = time = seconds = s
m = mass = Kilograms = kg
r = radius = distance = m
L = length = m
A = area = m^2
U = potential energy = J
K = kinetic energy = J
Φ_E = electric flux
$\vec{E}$ = electric field = $\frac{N}{C}$, $\frac{V}{m}$
σ = Surface charge density = $\frac{C}{SA(m^2)}$
ρ = volume charge density = $\frac{C}{V(m^3)}$
λ = linear charge density = $\frac{C}{L(m)}$
V = volt = $\frac{J}{C}$
I = current = Amperes = A = $\frac{C}{s}$
P = power = watts = W = $\frac{J}{s}$
R = resistance = ohm = Ω = $\frac{V}{A}$
C = capacitance = Farad = C = $\frac{C}{V}$
T = magnetic field = Tesla = $\frac{kg}{Cs}$
$\mu_0 = 4\pi\times10^{-7}$
Φ_B = magnetic flux = Weber = wb
L = Inductance = Henry = H
N = number of turns
n = number of turns per distance = $\frac{N}{m}$
T = torque = $\frac{Kgm^2}{s^2}$
τ = time constant = $\frac{L}{R}$

General stuff is black
Stuff from previous courses is Orange
Electric Field Equations/units are Green
Circuits are cyan
Magnets are purple
Blue is random info
Gold is diagrams
Magneto is notation stuff

SI prefixes
pico = 10^{-12} = p
nano = 10^{-9} = n
micro = 10^{-6} = μ
milli = 10^{-3} = m
centi = 10^{-2} = c
kilo = 10^3 = k
mega = 10^6 = M
giga = 10^9 = g

Notation

$\hat{r}$ = unit vector
$\vec{r}$ = Vector
$||r||$ = magnitude of vector
$\vec{r}_{12}$ = vector from 1 to 2
$\hat{i}$ = x unit vector = $\hat{x}$
$\hat{j}$ = y unit vector = $\hat{y}$
$\hat{k}$ = z unit vector = $\hat{z}$
E = electric field
$\mathcal{E}$ = EMF
ε = epsilon = constant usually
Φ_E = electric flux
Φ_B = magnetic flux
Σ = sum
R = radius of object
r = radius at point
• = dot product
x = cross product
>> = much greater than
<< = much less than
red text = Use # 2 of letter/symbol

$V_{sphere} = \frac{4}{3}\pi r^3$
$S.A._{sphere} = 4\pi r^2$
$A_{circle} = \pi r^2$

$\vec{F} = q\vec{E}$ Total force is sum of all force vectors. Make sure to include negatives!!
$\vec{F} = m\vec{a}$
$\vec{a} = \frac{m}{q}\vec{E}$
$\vec{E} = K_e\frac{q}{||\vec{r}||^2}\hat{r}$
$\vec{F}_{12} = K_e\frac{|q_1q_2|}{||\vec{r}_{12}||^2}\hat{r}_{12}$ Force between point charges; repulsive if signs are same, attractive if signs are opposite

$\vec{E}_{rod\ of\ charges} = K_e\frac{Q}{r\sqrt{r^2+z^2}}\hat{z}$ → z = length of rod, Q = total charge, r = distance to center of rod
$\vec{E}_{line\ of\ charge} = \frac{\lambda}{2\pi\epsilon_0 r}\hat{r}$
$\vec{E}_{cylindrical\ charge} = \frac{\rho R^2}{2\epsilon_0 r}\hat{r}$
$\vec{E}_{plane\ of\ charge} = \frac{\sigma}{2\epsilon_0}(\pm\hat{z})$
$\vec{E}_{outside\ gaussian\ sphere} = \frac{q}{4\pi\epsilon_0 R^2}\hat{r}$
$\vec{E}_{inside\ uniformly\ charged\ GS} = \frac{q_{in}r}{4\pi\epsilon_0 R^3}\hat{r}$
$\vec{E}_{inside\ G.S.} = 0$

Statics: $\vec{E}$ in conductors is ALWAYS 0.
$\vec{E} = -\nabla V$
$\vec{E} = \langle\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\rangle$
G. Spheres: R = of sphere, r = radius of gaussian sphere

Electric Dipoles

$\vec{E}_{dipole\ on\ axis} = K_e\frac{2q_d\vec{z}}{(\vec{z}-\frac{d}{2})^2(\vec{z}+\frac{d}{2})^2}\hat{z}$ → If $z >> d$, $\vec{E} = K_e\frac{qd}{z^3}\hat{z}$
d = distance between charges on dipole
z = distance to test charge from dipole axis
If a dipole is in a constant E field, net Force is 0.
However there will be torque to align the dipole with the field
$\vec{F}_{dipole} = q\vec{E}_{dipole}$ $V_{dipole} = \frac{qd}{r^2}$ $\vec{T} = \vec{r}\times\vec{F}$, $\vec{T} = \vec{p}\times\vec{E}$

$||\vec{E}|| = \frac{\Delta V}{d}$

Flux:

$\vec{A}$ = area w/ direction perpendicular to surface
$\Phi_E = EA\cos\theta$
$\Phi_E = \vec{E}\cdot\vec{A}$
θ is angle b/w $\vec{E}$ & $\vec{A}$
$\Phi_{E\ over\ surface} = \frac{q}{\epsilon_0}$
Net $\Phi_E = \frac{net\ q}{\epsilon_0}$

Potential:

V: volts
$\Delta V = -\vec{E}\cdot\vec{dr}$
$\Delta V = -E\Delta r\cos\theta$
$\Delta V = \mathcal{E}L$

Energy: Work is positive if two like charges are brought together

$K = \frac{1}{2}mv^2$ $U = K_e\frac{q_1q_2}{r}$ $W = \Delta U + \Delta K$ $\Delta U = q_t\Delta V$
$U_{min} = -p\mathcal{E}$ $U = -\int\vec{F}\cdot d\theta$ $U = -\vec{p}\cdot\vec{E}$ (dot product of dipole moment and E field)
$U_{max} = p\mathcal{E}$ $U_{total} = \sum U = \sum K_e\frac{q_iq_j}{r_{ij}}$ (sum of all charge pairs)
Max change in $U = 2p\mathcal{E}$
$\Delta U = -e\Delta V \rightarrow \Delta K$

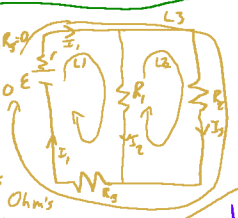
Kirchhoff's Rules

- current flows "downhill"

Junction Rule: $\sum$ all currents in = $\sum$ all currents out
$I_1 = I_2 + I_3$
$I_2 + I_3 = I_1$

Loop Rule
Loop 1: $\mathcal{E} - I_1r - I_2R_1 - I_1R_3 = 0$
Loop 2: $+I_2\mathcal{E}_1 - I_3R_2 = 0$
Loop 3: $\mathcal{E} - I_1r - I_2R_3 - I_3R_2 = 0$

Once all I's are known, ΔV is easy from Ohm's Law



- Treat battery as + $\mathcal{E}$ looping from − to +
- Batteries opposing each other work against each other

Circuit Stuff

V_d = drift speed of electrons
Δt_c = time between electron collisions
n_e = # of free electrons per volume
$\vec{J}$ = current density
σ = conductivity
ρ = resistivity
r = internal resistance
$\mathcal{E}$ = EMF = energy from power source

$\rho = \frac{1}{\sigma}$
$\vec{J} = \sigma\vec{E}$
$\vec{J} = \frac{I}{A}(-\hat{v}_d)$

Ohm's Law: $\Delta V = IR$
ΔV in battery = $\mathcal{E} - Ir$
$C = k\frac{Q}{\Delta V} = k\frac{\epsilon_0 A}{d}$
$C_{sphere} = k\frac{R_1R_2}{k_e(R_2-R_1)}$
$C = k\frac{q_d}{AE_0}$
$R = \frac{\rho L}{A}$
$P = I^2R = I\Delta V = \frac{\Delta V^2}{R}$
$U_{cap} = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}\frac{Q^2}{C}$

$V_d = \frac{1}{2}\frac{e\vec{E}}{m_e}\Delta t_c$
$a = -\frac{e\vec{E}}{m_e}$
$L = v_d(t)$
$I = q n_e A v_d$
$I = \frac{\sigma A\Delta V}{L}$

Resistors: Series $R_1 + R_2 + R_3 \dots$; Parallel $\frac{1}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\dots}$
Capacitors: Parallel $C_1 + C_2 + C_3 \dots$; Series $\frac{1}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}\dots}$
Dielectric = k Multiplier of Capacitor

Magnetics!

$q\vec{v}\times\vec{B} = \vec{F}_{mag}$
$F_{mag} = qvBsin\theta$
$F = IBL$
$\vec{F}_{mag} = \int I\vec{dL}\times\vec{B}$
$F_{net} = \sum$ all forces

If $v \perp B$: $qvB = \frac{mv^2}{r}$ → $r = \frac{mv}{qB}$
North pole = + → Lines go in
South pole = − → Lines go out
$B_{center\ of\ loop} = \frac{\mu_0}{2\pi}\frac{I}{R}$ = line of current

$\vec{B}$ = magnetic field vector
$B_{on\ axis} = \frac{\mu_0}{4\pi}\frac{2NIR}{(r^2+z^2)^{3/2}}(-\hat{k}) \overset{z>>r}{\rightarrow} \frac{\mu_0 NIR^2}{2|z|^3}\hat{k}$

Torque: $\vec{T} = \vec{r}\times\vec{F}$ $\vec{T} = rFsin\theta$
$\vec{T}_{net} = NIAB = \vec{\mu}_m\times\vec{B}$

$W = -\Delta U$
$U = -\vec{\mu}_m\cdot\vec{B}$

cyclotron frequency: $\omega = \frac{qB}{m}$
Frequency = $\frac{1}{T} = \frac{qB}{2\pi m}$
CW (2) rotation is −
CCW (G) rotation is +

Magnetic Dipoles

Dipole moment = $\mu_m = NIA$
$z >> R$: $(\frac{\mu_0}{4\pi})\frac{2\mu_m}{|z|^3} = B_{mag\ dipole\ on\ axis}$
$B_{mag\ dipole,\perp axis} = \frac{\mu_0}{4\pi}\frac{\mu_m}{|r|^3}$ if $z >> r$

$B_{solenoid} = \mu_0 nI$, direction given by RHR

$\hat{\phi}$ = unit vector in direction of current given by RHR
$I = \frac{\mathcal{E}}{R}$ $\mathcal{E} = \mathcal{E}MF$ $\mathcal{E}_{induced} = N(-\frac{d}{dt}\Phi_B)$

Ampere's Law: $\vec{B} = \frac{\mu_0}{2\pi}\frac{I}{r}\hat{\phi}$
Φ_B = magnetic flux = $NA\mu_0 n_s I$
L = inductance = Henry = $-N\mu_0 n_s A$
M = mutual inductance = $\mu_0 n_s A$
$\mathcal{E} = -M\frac{dI}{dt}$
$\mathcal{E}_{back} = -N\mu_0 N_s A\frac{dI}{dt}$
$\oint\vec{E}\cdot\vec{dl} = -N\frac{d\Phi_B}{dt}$
$U = \frac{1}{2}LI^2$

RL/LC/RLC circuits
$I(t) = \frac{\mathcal{E}}{R}(1-e^{-\frac{t}{\tau}})$ $I(t) = I_0e^{-\frac{t}{\tau}}$
$\mathcal{E} - IR - L\frac{dI}{dt} = 0$
$\tau = \frac{L}{R}$
$I_0 = \frac{1}{\sqrt{LC}}q_0$
$q_0 = CV$

Oscillating circuit (LC):
$\frac{1}{\sqrt{LC}}$ = frequency → hz = $\frac{1}{2\pi\sqrt{LC}}$

c = speed of light = 3×10^8 m/s = $\frac{1}{\sqrt{\mu_0\epsilon_0}}$

Gauss's Law: $\oint\vec{E}\cdot\vec{dA} = \frac{q_{enclosed}}{\epsilon_0}$
Gauss's Law, mag fields: $\oint\vec{B}\cdot\vec{dA} = 0$
Maxwell-Ampere Law: $\oint\vec{B}\cdot\vec{dL} = \mu_0 I_{enc} + \mu_0\epsilon_0\frac{d}{dt}\int\vec{E}\cdot\vec{dA}$
Faraday's Law: $\oint\vec{E}\cdot\vec{dL} = -\frac{d}{dt}\int\vec{B}\cdot\vec{dA}$

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